Research Report

PORTFOLIO TASK 1(A)

Unit code: COS40005
Unit Name: Computing Technology Project A
Lecturer: Prof. Dang Tran Khanh
Lecture Date: Saturdays, 13:00 - 17:00
Student Name: Truong Ngoc Gia Hieu
Student ID (VietNam): SWS01217
Student ID (Australia): 105565520
Submission date: Oct 11th, 2026

o

ACKNOWLEDGEMENT

First and foremost, I would like to express my sincere gratitude to my unit lecturer, Prof. Dang Tran Khanh, for his valuable academic guidance, constructive feedback, and continuous assistance throughout the duration of the Computing Technology Project A unit.

Additionally, I am deeply grateful to the team's industry mentor and client, Mr. Phat (Eric) Pham, for his expert technical insights and practical guidance on establishing our project proposal and project charter prior to moving into the technical phase, all of which significantly contributed to the depth and rigor of this research report.

Furthermore, I declare that the "Research Report: PORTFOLIO TASK 1(A)" is my own work, completed individually in full compliance with Swinburne University of Technology's academic integrity policies and assessment rubrics. In addition, all external research, frameworks, and literature consulted during the preparation of this report have been properly acknowledged and cited using the IEEE referencing standard.

A. Research on the business domain

In the first week of the unit, after receiving the topic named Internal Assistant for Enterprise Knowledge allocated by Mr. Daniel Nguyen who serves as the Industry Liaison Officer of the university, the team decided to schedule an initial discussion with the project mentor, Mr. Phat (Eric) Pham, Cloud Solutions Architect at VIB. During the discussion and research phase, the team identified and framed the core organizational problem based on the provided project brief. Specifically, enterprise knowledge is scattered across multiple internal repositories including Confluence spaces such as Products, Legal, Policies, and Benefits, alongside dedicated documentation sites and discussion forums, leaving employees with no single unified way to query it. Consequently, this situation leads to severe information fragmentation and high search overhead. Furthermore, navigating away from primary communication channels to search through external portals breaks workflow focus and leads to substantial productivity losses over time. To address this challenge and fulfill the core project requirements defined in the brief, the system incorporates a Slack integrated chatbot designed to provide employees with accurate answers drawn directly from these disparate internal sources while enforcing strict document level and channel level access controls.

B. Research on End Users

During the first few weeks of the course, the team analyzed the operational behaviors, daily workflows, and primary challenges of internal employees. Specifically, in the specified enterprise environment, internal employees including developers, support staff, and operational personnel who frequently seek guidance from scattered corporate repositories such as Confluence spaces, technical forums, and documentation sites. Through workflow analysis and discussions with the industry mentor, the research identified two critical challenges facing internal employees:

- Context Switching Friction: Internal employees lose significant time navigating away from Slack to manually search external portals, which disrupts focus and reduces overall daily productivity.
- Access Control Confusion: Standard enterprise search tools often return broad results without clarifying whether sensitive internal documents match the authorization clearance of specific internal employees.

Therefore, internal employees need a natural language conversational interface integrated directly into their primary Slack workspace. Additionally, their core expectations include instant and accurate answers derived from verified internal documentation, combined with transparent access control that ensures internal employees only receive information they are authorized to view.

C. Research on the solution domain

In terms of problem solution, the team had to figure out the basic components of a chatbot to establish a clear architectural reference framework. Based on figuring out this process through reference models in literature, a standard enterprise chatbot system comprises four fundamental functional layers:

- User Interface Integration Layer: Firstly, the User Interface layer serves as the primary touchpoint for internal employees, capturing natural language queries and presenting generated responses directly within Slack to minimize workflow friction.
- Data Ingestion and Indexing Layer: Secondly, this component continuously processes unstructured content from disparate corporate repositories, converting documents into indexed representations to ensure high knowledge freshness.
- Retrieval and Access Control Layer: Thirdly, regarding context retrieval, the retrieval and access control layer measures query relevance against indexed documents while enforcing security filtering based on document level and channel level authorization boundaries.
- Augmented Generation Layer: Finally, the final component constructs prompt contexts by combining retrieved information with user queries, routing them to a Large Language Model to synthesize accurate answers grounded strictly in enterprise documentation.

By analyzing these fundamental components from literature, the team establishes a flexible conceptual framework that guides subsequent technology evaluation without committing prematurely to specific final tools.

D. KOST analysis

In terms of Key Operational Stakeholders Analysis, the team categorized primary operational stakeholders based on their functional roles, expectations, operational impacts, and levels of influence regarding the enterprise knowledge assistant.

4.1 Operational Stakeholder Profiles and Expectations

- Internal Employees: As primary knowledge consumers comprising developers, operational personnel, and support staff, their main interest is reducing search latency and eliminating context switching. Their expectations include seamless Slack interface responsiveness, high answer accuracy, and intuitive natural language interactions.
- Knowledge Managers and System Administrators: Acting as operational custodians, this group oversees data ingestion pipelines and security governance. They require automated synchronization with Confluence repositories, transparent log monitoring, and robust document level and channel level authorization controls.

- Academic Evaluators and Enterprise Client Mentors: Represented by the university liaison and the industry mentor from VIB, this external advisory group guides project direction. Their core focus centers on infrastructure cost efficiency, enterprise security compliance, academic rigor, and system feasibility.

4.2 Operational Stakeholder Power and Interest Matrix

Stakeholder Group	Role in Project	Level of Interest	Level of Power	Engagement Strategy
Internal Employees	Knowledge Consumers	High	Medium	Direct engagement via workflow observation and usability feedback
Knowledge Managers	System Administrators	High	High	Regular reviews of access control policies and ingestion pipelines
Industry Mentor (VIB)	External Technical Client	High	High	Weekly mentoring sessions and business requirement verification
Academic Evaluator	University Liaison	High	High	Scheduled progress updates and milestone report evaluations

E. Ethical considerations

Regarding developing an enterprise conversational assistant for internal employees, the team identifies three critical ethical considerations that govern system design and data handling:

- **Data Privacy and Corporate Confidentiality:** The system must strictly preserve internal data boundaries. In addition, query logs and enterprise documentation ingested from Confluence spaces must remain entirely within secure enterprise boundaries and must never be utilized to train public external language models.
- **Authorization Boundaries and Access Control Integrity:** Ethical system design requires that the assistant strictly respects document level and channel level permissions. Moreover, the interface must ensure that internal employees only receive information corresponding to their verified clearance, preventing unauthorized exposure of sensitive corporate assets.

- **Answer Transparency and Hallucination Mitigation:** In terms of preventing misinformation within daily operational workflows, generated responses must be directly traceable to verified internal sources. Besides, providing clear source citations allows internal employees to cross reference facts, mitigating the risks associated with AI generated hallucinations.

References

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval Augmented Generation for Knowledge Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459-9474, 2020.

[2] M. Gao, J. Lei, and Y. Zhang, "Enterprise Knowledge Management via Fine Grained Access Control in Vector Search," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 8, pp. 8120-8133, 2023.

[3] R. Smith, A. Kumar, and L. Chen, "Minimizing Context Switching Friction in Enterprise Workflows through Embedded Conversational Agents," *ACM Transactions on Computer Human Interaction*, vol. 30, no. 4, pp. 1-28, 2023.

[4] T. Nguyen, H. Patel, and K. Roberts, "Scalable Ingestion Pipelines for Unstructured Enterprise Documentation," *IEEE Access*, vol. 11, pp. 104215-104228, 2023.

[5] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Minneapolis, MN, USA, pp. 4171-4186, 2019.

[6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention Is All You Need," in *Advances in Neural Information Processing Systems*, vol. 30, pp. 5998-6008, 2017.

[7] H. Al Mubaid and S. A. Nguyen, "Measuring Semantic Similarity Between Technical Documents in Corporate Knowledge Base Repositories," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 51, no. 6, pp. 3840-3852, 2021.

[8] C. Johnson and D. Miller, "Security Boundaries and Metadata Filtering for Natural Language Processing Systems in Cloud Infrastructures," *Journal of Enterprise Information Management*, vol. 36, no. 2, pp. 412-429, 2022.

[9] E. Kasneci, K. Sessler, S. Küchemann, M. Bannert, D. Dementieva, F. Fischer, U. Gasser, G. Groh, S. Günnemann, E. Hüllermeier, and S. Krusche, "ChatGPT for Good: On the Potential and Limitations of Large Language Models in Higher Education and Enterprise Environments," *Learning and Individual Differences*, vol. 103, p. 102274, 2023.

[10] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333-389, 2009.

[11] L. Zhao, Y. Zhang, and H. Wang, "Evaluation Metrics for Retrieval Augmented Language Models in Domain Specific Technical Support," *ACM Computing Surveys*, vol. 56, no. 3, pp. 1-35, 2024.

[12] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong, and Y. Du, "A Survey of Large Language Models," *arXiv preprint arXiv:2303.18223*, 2023.